

CyberSecurity Lab1

Part 1 : understanding Password Security

Step1 : Analyse Weak Passwords

Discussion

1. Why are these passwords insecure?

Passwords such as 123456, password, admin, 12345678, and qwerty are insecure because they are very common and easy to guess. They are short and do not contain a strong mix of uppercase letters, lowercase letters, numbers, and special characters. Because many people use them, attackers can easily guess them using automated tools.

2. How quickly could a computer guess these passwords?

A computer can guess these passwords very quickly, sometimes in a few seconds. Attackers use automated tools that test thousands or even millions of password combinations in a short time. Since these passwords are common, they are usually cracked almost immediately.

3. Attackers often use automated tools to try many passwords quickly. A dictionary attack tests a list of common passwords, while a brute force attack tries many possible combinations until the correct password is found. These methods make weak passwords easy to crack.

Step 2: Create Strong Passwords

Let's write my examples below:

1. Cyb3r\$\$hield_92
2. Str0ng!P@ssKey45
3. Gi\$ecur3_Lab88

Part 2: Observing Network Traffic

Step 1: Launch Wireshark

Step 2: Generate Network Traffic

Step 3: Stop Packet Capture

Part 3: Packet Analysis

Step 1: DNS Traffic

Let's write one domain name observed:

www.google.com

No.	Time	Source	Destination	Protocol	Length	Info
9670	118.012524	192.168.100.121	192.168.100.1	DNS	79	Standard query 0xe474 A accounts.google.com
9673	118.030986	192.168.100.1	192.168.100.121	DNS	90	Standard query response 0x741e A www.google.com A 172.217.171.36
9674	118.030986	192.168.100.1	192.168.100.121	DNS	343	Standard query response 0x1dbb A accounts.google.com A 173.194.76.84 NS ns4.google.com NS ...
9675	118.030986	192.168.100.1	192.168.100.121	DNS	129	Standard query response 0xafec HTTPS accounts.google.com SOA ns1.google.com
9676	118.054612	192.168.100.1	192.168.100.121	DNS	95	Standard query response 0xe474 A accounts.google.com A 173.194.76.84
9824	119.103322	192.168.100.121	192.168.100.1	DNS	74	Standard query 0x2c9c A www.google.com
9825	119.103388	192.168.100.121	192.168.100.1	DNS	74	Standard query 0x57c2 A ogs.google.com
9826	119.106837	192.168.100.121	192.168.100.1	DNS	74	Standard query 0xdf9 HTTPS ogs.google.com
9827	119.107522	192.168.100.121	192.168.100.1	DNS	74	Standard query 0xadf3 A ogs.google.com
9828	119.213069	192.168.100.1	192.168.100.121	DNS	90	Standard query response 0x2c9c A www.google.com A 172.217.171.36
9829	119.213069	192.168.100.1	192.168.100.121	DNS	359	Standard query response 0x57c2 A ogs.google.com CNAME ww3.l.google.com A 142.251.142.142 ...
9830	119.213069	192.168.100.1	192.168.100.121	DNS	359	Standard query response 0xadf3 A ogs.google.com CNAME ww3.l.google.com A 142.251.142.142 ...
9831	119.213069	192.168.100.1	192.168.100.121	DNS	145	Standard query response 0xdf9 HTTPS ogs.google.com CNAME ww3.l.google.com SOA ns1.google...
9862	119.725829	192.168.100.121	192.168.100.1	DNS	75	Standard query 0x086f HTTPS ssl.gstatic.com
9863	119.726813	192.168.100.121	192.168.100.1	DNS	75	Standard query 0x9dc2 A ssl.gstatic.com

Step 2: HTTP Traffic

No.	Time	Source	Destination	Protocol	Length	Info
http2	590645	192.168.100.121	142.250.74.67	HTTP	261	GET /gsr1/gsr1.crl HTTP/1.1
http3	7906185	142.250.74.67	192.168.100.121	HTTP	280	HTTP/1.1 304 Not Modified
	2776	24.865274	192.168.100.121	HTTP	256	GET /r/gsr1.crl HTTP/1.1
	2779	24.986492	192.168.100.121	HTTP	281	HTTP/1.1 304 Not Modified
	2781	25.026274	192.168.100.121	HTTP	254	GET /r/r4.crl HTTP/1.1
	2782	25.144705	192.168.100.121	HTTP	280	HTTP/1.1 304 Not Modified
	2783	25.165566	192.168.100.121	HTTP	254	GET /r/r1.crl HTTP/1.1
	2784	25.285628	192.168.100.121	HTTP	280	HTTP/1.1 304 Not Modified
	2791	25.450267	192.168.100.121	HTTP	281	GET / HTTP/1.1
	2793	25.557369	184.28.84.204	HTTP	317	HTTP/1.1 304 Not Modified
	2800	25.744311	192.168.100.121	HTTP	317	GET /DigiCertHighAssuranceEVRootCA.crl HTTP/1.1
	2803	25.844144	23.207.17.2	HTTP	285	HTTP/1.1 304 Not Modified
	2804	25.872942	192.168.100.121	HTTP	307	GET /DigiCertGlobalRootG2.crl HTTP/1.1
	2807	25.967840	23.207.17.2	HTTP	284	HTTP/1.1 304 Not Modified
	2808	26.006618	192.168.100.121	HTTP	307	GET /DigiCertGlobalRootG3.crl HTTP/1.1
	2810	26.100313	23.207.17.2	HTTP	285	HTTP/1.1 304 Not Modified
	7499	90.024738	192.168.100.121	HTTP	136	GET /ncc.txt HTTP/1.1
	7501	90.137895	2.19.251.139	HTTP	205	HTTP/1.1 200 OK (text/html)

> Frame 7501: Packet, 205 bytes on wire (1640 bits), 205 bytes captured (1640 bits) on interface 0

> Ethernet II, Src: HuaweiTechno_38:6a:5f (48:4c:29:38:6a:5f), Dst: Intel_a5:c5:21 (f0:c2:95:55:a5:c5)

> Internet Protocol Version 4, Src: 2.19.251.139, Dst: 192.168.100.121

> Transmission Control Protocol, Src Port: 80, Dst Port: 1261, Seq: 1, Ack: 83, Len: 151

> Hypertext Transfer Protocol

> Line-based text data: text/html (1 lines)

0000 f0 d5 bf a5 c5 21 48 4c 29 38 6a 5f 08 00 45 00IHL)8J...E

0010 00 bf 6f ff 40 00 32 06 b5 79 02 13 fb 8b c0 a8 ...o@2...y....

0020 64 79 00 50 04 ed fc a4 a5 67 f8 33 41 43 50 18 dy p...g 3ACP

0030 01 f6 4f 15 00 00 48 54 54 50 2f 31 2e 31 20 32 ...O...HT TP/1.1 2

0040 30 30 20 4f 4b 0d 0a 43 6f 6e 74 65 6e 74 2d 54 00 OK - Content-T

0050 79 70 65 3a 20 74 65 78 74 2f 68 74 6d 6c 0d 0a ype: tex t/html-

0060 43 6f 6e 74 65 6e 74 2d 4c 65 6e 67 74 68 3a 20 Content- Length:

0070 32 36 00 0a 44 61 74 65 3a 20 4d 6f 6e 2c 20 31 26 Date : Mon, 1

0080 36 20 4d 61 72 20 32 30 32 36 20 31 37 3a 32 37 6 Mar 20 26 17:27

0090 3a 33 35 20 47 4d 54 0d 0a 43 6f 6e 6e 65 63 74 :35 GMT- Connect

00a0 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 6d ion: kee p-alive

00b0 0a 0d 0a 4e 65 74 77 6f 72 6b 20 43 6f 6e 6e 65 - Netwo rk Conne

00c0 63 74 69 76 69 74 79 20 43 68 65 63 6b ctivity Check

Step 3: Encrypted Traffic

The image shows a Wireshark network traffic capture. The top pane displays a list of 19 packets, all of which are TLSv1.3 Application Data. The source IP is 192.0.66.2 and the destination IP is 192.168.100.121. The bottom pane shows a detailed view of the selected packet (No. 19), which is a TLSv1.3 Application Data packet. It displays the raw data in hexadecimal and ASCII, along with various fields like Sequence Number, Acknowledgment Number, and Stream Index.

No.	Time	Source	Destination	Protocol	Length	Info
5724	85.729033	192.0.66.2	192.168.100.121	TLSv1.3	12762	Application Data, Application Data, Application Data, Application Data, Application Data, ...
5726	85.729921	192.0.66.2	192.168.100.121	TLSv1.3	11350	Application Data, Application Data, Application Data, Application Data
5737	85.755197	192.0.66.2	192.168.100.121	TLSv1.3	4290	Application Data
5739	85.756706	192.0.66.2	192.168.100.121	TLSv1.3	16998	Application Data, Application Data, Application Data, Application Data
5740	85.756706	192.0.66.2	192.168.100.121	TLSv1.3	1434	Application Data, Application Data
5748	85.836852	192.168.100.121	192.0.66.2	TLSv1.3	161	Application Data
5752	85.945539	192.0.66.2	192.168.100.121	TLSv1.3	395	Application Data
5753	85.945539	192.0.66.2	192.168.100.121	TLSv1.3	513	Application Data
5754	85.945539	192.0.66.2	192.168.100.121	TLSv1.3	85	Application Data
5768	86.168064	192.168.100.121	172.217.20.106	TLSv1.3	373	Client Hello (SMI=jnn-pa.googleapis.com)
5779	86.304718	172.217.20.106	192.168.100.121	TLSv1.3	1470	Server Hello, Change Cipher Spec
5783	86.306417	172.217.20.106	192.168.100.121	TLSv1.3	177	Application Data
5785	86.313784	192.168.100.121	172.217.20.106	TLSv1.3	144	Change Cipher Spec, Application Data
5786	86.317194	192.168.100.121	172.217.20.106	TLSv1.3	146	Application Data
5787	86.317293	192.168.100.121	172.217.20.106	TLSv1.3	496	Application Data
5788	86.414896	192.168.100.121	192.0.66.2	TLSv1.3	192	Application Data
5789	86.435200	172.217.20.106	192.168.100.121	TLSv1.3	1062	Application Data, Application Data
5790	86.435426	172.217.20.106	192.168.100.121	TLSv1.3	89	Application Data

Source Port: 443
Destination Port: 1245
[Stream index: 41]
[Stream Packet Number: 1368]
[Conversation completeness: Complete, WITH_DATA (63)]
[TCP Segment Len: 5648]
Sequence Number: 6511184 (relative sequence number)
Sequence Number (raw): 1348391047
[Next Sequence Number: 6516832 (relative sequence number)]
Acknowledgment Number: 7225 (relative ack number)
Acknowledgment Number (raw): 353327220
0101 = Header Length: 20 bytes (5)

Answers to the questions

1. Approximately 1500 packets were captured during the experiment.
2. Let's list three protocols observed during the experiment:
DNS, HTTP, TLS
3. Encryption is important when browsing the internet because it protects sensitive information transmitted over the Internet. Without encryption, data such as passwords, personal information, or financial data could be intercepted and read by attackers.
4. Let's show the types of information could attackers obtain from unencrypted traffic:
An attacker could access several types of sensitive information, including login credentials, personal data, session cookies, email addresses, requests sent to websites, or even the pages visited by the user.

Lab Report

This experiment clearly shows that network traffic can be easily captured and analyzed with accessible tools. It also demonstrates that weak passwords represent a major vulnerability and that unencrypted communications can expose sensitive information. Therefore, the use of encryption and strong passwords is essential to protect users' security and privacy on the Internet.